

CRF Fluctuation Law

$$\sigma_N/\sqrt{N} = 1/\sqrt{n \cdot T_{\text{avg}}} \text{ at Canonical } \kappa$$

Derived at $\kappa = 0.15$, confirmed 5 seeds. κ -dependence identified as open problem.

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Abstract

We identify and derive a fluctuation law governing mass spread across nodes in the Constrained Reality Framework (CRF) driven regime, and establish its precise scope.

Main result (derived). At canonical coupling $\kappa = 0.15$, the normalised firing dispersion satisfies:

$$Q \equiv \frac{\sigma_N}{\sqrt{N}} = \frac{1}{\sqrt{n \cdot T_{\text{avg}}}} = 0.05941,$$

where $n = 8$ (SO(3), #16) and $T_{\text{avg}} = 35.42$ sweeps (#21a). Measured across 5 seeds: $Q = 0.05880 \pm 0.00040$ (CV 0.7%, gap 1.0%).

κ -dependence test (empirical). Varying $\kappa \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$ shows Q is not κ -independent. Empirically $Q \approx 0.103 \times \kappa^{0.274}$ ($R^2 = 0.95$), but this scaling is *not yet derived* from the δ -chain. Drift-diffusion analysis predicts the opposite sign ($Q \propto \kappa^{-1/6}$), indicating a collective network mechanism not captured by single-node theory.

Scope. The derived law holds at $\kappa = 0.15$. The κ -scaling is an empirical Hypothesis pointing toward an undiscovered coupling between κ , network topology, and firing-rate heterogeneity.

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1 Background

In CRF, each node i accumulates mass through R-events: $\Omega_i = \kappa \cdot N_{\text{eff},i}$, $\kappa = 0.15$. The spread of Ω across nodes characterises crystallisation uniformity.

Previous work (Experiment C) observed $\xi/\sigma_\Omega \approx 1$ in short runs. Extended runs showed σ_Ω grows with $\bar{\Omega}$ — not a fixed point. The correct invariant is the normalised ratio $Q = \sigma_N/\sqrt{\bar{N}}$, which is stationary throughout long runs.

2 The Derived Law at $\kappa = 0.15$

Fluctuation Law (canonical κ)

$$Q \equiv \frac{\sigma_N}{\sqrt{\bar{N}}} = \frac{1}{\sqrt{n \cdot T_{\text{avg}}}},$$

where $\sigma_N = \sigma_\Omega/\kappa$, $\bar{N} = \bar{\Omega}/\kappa$, $n = 8$ (SO(3)), $T_{\text{avg}} = 2nK^*/[(n-1)\varepsilon_0] = 35.42$ (#21a).

Derivation sketch. Between R-events, n symmetry modes each accumulate ε_0 per sweep. Collective firing requires all n modes to cross threshold simultaneously, suppressing variance by $1/n$ relative to independent Poisson. Over T_{avg} sweeps the normalised dispersion is $1/\sqrt{nT_{\text{avg}}}$.

Numerical result

$$Q = \frac{1}{\sqrt{8 \times 35.42}} = 0.05941.$$

Measured (5 seeds, 1500 sweeps, $\kappa = 0.15$): $Q = 0.05880 \pm 0.00040$. Gap: 1.0%.

2.1 Multi-seed confirmation

Seed	Q (mean)	Q (std)	Late-run slope
42	0.05831	0.00124	$+5.6 \times 10^{-6}$
123	0.05835	0.00161	-1.2×10^{-5}
777	0.05915	0.00179	-1.6×10^{-5}
2024	0.05893	0.00157	-4.1×10^{-6}
99	0.05928	0.00154	-2.9×10^{-5}
Mean	0.05880	0.00040	all slope $< 10^{-4}$

The variance suppression factor $\sqrt{nT_{\text{avg}}} \approx 16.8$ matches the collapse premium $\beta/\delta\bar{H} \approx 19$ from CRF State Summary v4 at the 11% level, suggesting a deeper connection (recorded as open below).

3 κ -Dependence Test

3.1 Protocol

CRF Hybrid V20.5, $N = 500$, seed = 42, 1200 sweeps. $\kappa \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$. Q computed from the last 40% of snapshots.

3.2 Results

κ	Q	T_{avg} (sw)	$Q \cdot \sqrt{nT_{\text{avg}}}$	pred. (Wald)
0.05	0.0475	78.1	1.187	0.0594
0.10	0.0521	69.7	1.229	0.0594
0.15	0.0591	66.4	1.361	0.0594
0.20	0.0671	64.8	1.526	0.0594
0.30	0.0769	63.7	1.736	0.0594

Three findings:

- (1) Q increases with κ : empirically $Q \approx 0.103 \times \kappa^{0.274}$, $R^2 = 0.95$, CV = 3.9% across the five κ values.
- (2) $Q \cdot \sqrt{nT_{\text{avg}}}$ is not constant (ranges 1.19 to 1.74), ruling out the simple extension $Q = 1/\sqrt{nT_{\text{avg}}(\kappa)}$.
- (3) T_{avg} itself changes with κ ($\propto \kappa^{-0.116}$), confirming that the Wald formula (#21a) — derived at $\kappa = 0.15$ with the initial threshold K^* — does not generalise directly to other κ .

3.3 Drift-diffusion analysis and the sign problem

Single-node drift-diffusion predicts:

$$Q_{\text{dd}} = \frac{\sigma_{\text{DKL}}}{\mu \sqrt{K^* \sqrt{M}}} \propto \kappa^{-1/6},$$

where $M = 1 + \kappa \bar{N}$ and $\sigma_{\text{DKL}}/\mu = 1.51$. This gives $Q \sim 3\text{--}4$ (factor 50–70× too large) and the *wrong sign*: theory predicts Q decreases with κ , simulation shows Q increases.

The sign contradiction rules out single-node mechanisms. $Q(\kappa)$ reflects a collective network effect — likely firing-rate heterogeneity across nodes, modulated by the mass-threshold coupling $\theta_{\text{eff}} = K^* \sqrt{1 + \kappa \bar{N}_{\text{eff}}}$.

4 Status Table

Result	Status	Note
$Q = 1/\sqrt{nT_{\text{avg}}}$ at $\kappa = 0.15$	Derived	5 seeds, 1.0% gap
Q stationary (slope $< 10^{-4}$)	Confirmed	all 5 seeds
Variance suppression $\approx 17\times$	Confirmed	$\sqrt{nT_{\text{avg}}} = 16.8$
Q κ -independent	Falsified	Priority-2 test
$Q = 1/\sqrt{nT(\kappa)}$ with measured T	Falsified	product $\neq 1$
$Q \approx 0.103 \times \kappa^{0.274}$	Hypothesis	$R^2 = 0.95$, not derived
Sign of $dQ/d\kappa$ from drift-diffusion	Falsified	predicts opposite sign
Q from collective network effect	Hypothesis	rules out single-node
Exponent 0.274 from δ -chain	Open	key next target
$\sqrt{nT} \approx \beta/\delta\bar{H}$ connection	Hypothesis	11% gap at $\kappa = 0.15$

5 Open Questions

Priority 1: Derive $Q(\kappa)$ from collective firing statistics

The empirical scaling $Q \propto \kappa^{0.274}$ and the sign failure of drift-diffusion both point to a collective mechanism. The correct framework is firing-rate heterogeneity across nodes: $\text{Var}(N_i) = \bar{N} + s^2 \text{Var}(\lambda_i)$, where λ_i is node i 's firing rate. Stationarity of Q requires $\text{Var}(\lambda_i) \propto 1/s$, consistent with mass-driven slowdown. Deriving how $\text{Var}(\lambda_i)$ depends on κ through $\theta_{\text{eff}} = K^* \sqrt{1 + \kappa N_{\text{eff}}}$ is the key open step.

Priority 2: $T_{\text{avg}}(\kappa)$ self-consistent equation

Extend Wald formula (#21a) to include mass-threshold coupling. The self-consistent equation $T = K^* \sqrt{1 + \kappa s/T} / \mu$ gives $T \propto (\kappa s)^{1/3}$ at large s , but measured $T_{\text{avg}} \propto \kappa^{-0.116}$ at fixed sweep — the two scalings are inconsistent, suggesting a finite- s correction is needed.

Priority 3: Connection to collapse premium

$\sqrt{nT_{\text{avg}}} \approx 17$ and $\beta/\delta\bar{H} \approx 19$ at $\kappa = 0.15$. Does this ratio stay near 17–19 at other κ ? If yes: the connection is κ -independent even if Q is not, pointing to a deeper identity in the δ -chain.

*The law held. Then the boundary spoke.
What fails cleanly is as informative as what holds.
The scope of the law is now known.
The mechanism beyond it is the next question.*